ENLARGING THE ARTINIAN APPROXIMATION RING TO CONTAIN A POWER OF A GIVEN UNIT

THE STACKS PROJECT AUTHORS

Editorial note. Input lemma-solution-modulo is a published entry (tag 07FH) used as a standing input by its display number; not proven on the author's behalf.

Allowed standard prerequisites (editorial).

- Artinian approximation factorization. lemma-solution-modulo (same chapter, PUBLISHED 07FH). Supplies the input factorization k[y]_p/p^n -> D -> bar-Lambda being enlarged.
- Structure of algebraic field extensions in characteristic p. Fields, section algebraic (cross-chapter, named). A p-power of an algebraic element becomes separably algebraic.
- Henselianness of Artinian local rings. Algebra, lemma local-dimension-zero-henselian (cross-chapter, named). D and bar-Lambda are henselian, enabling the finite etale lifting.
- Equivalence of finite etale extensions and separable residue extensions. Algebra, lemma henselian-cat-finite-etale (cross-chapter, named). Produces D -> D' with residue field F(alpha^q)/F.

Source reference. Original reference 04GK: focused-seeds/geometry/S9-smo-enlarge-solution-modulo/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06RS: focused-seeds/geometry/S9-smo-enlarge-solution-modulo/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 037H: focused-seeds/geometry/S9-smo-enlarge-solution-modulo/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 07FH: sources/stacks-smoothing/pinned-original-source.tex, source lines 2769–2769. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:smoothing.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let $\varphi : k[y_1, \ldots, y_m] \to \Lambda$, n, $\mathfrak{q}$, $\mathfrak{p}$ and*

$$k[y_1, \ldots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \to D \to \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$$

be as in Lemma 07FH. Then for any $\lambda \in \Lambda \setminus \mathfrak{q}$ there exists an integer $q > 0$ and a factorization

$$k[y_1, \ldots, y_m]_{\mathfrak{p}}/\mathfrak{p}^n \to D \to D' \to \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$$

such that $D \to D'$ is an essentially smooth map of local Artinian rings, the last arrow is flat, and λ^q is in D'.

Proof. Set $\bar{\Lambda} = \Lambda_{\mathfrak{q}}/\mathfrak{q}^n\Lambda_{\mathfrak{q}}$. Let $\bar{\lambda}$ be the image of λ in $\bar{\Lambda}$. Let $\alpha \in \kappa(\mathfrak{q})$ be the image of λ in the residue field. Let $k \subset F \subset \kappa(\mathfrak{q})$ be the residue field of D. If α is in F then we can find an $x \in D$ such that $x\bar{\lambda} = 1 \bmod \mathfrak{q}$. Hence $(x\bar{\lambda})^q = 1 \bmod (\mathfrak{q})^q$ if q is divisible by p. Hence $\bar{\lambda}^q$ is in D. If α is transcendental over F, then we can take $D' = (D[\bar{\lambda}])_{\mathfrak{m}}$ equal to the subring generated by D and $\bar{\lambda}$ localized at $\mathfrak{m} = D[\bar{\lambda}] \cap \mathfrak{q}\bar{\Lambda}$. This works because $D[\bar{\lambda}]$ is in fact a polynomial algebra over D in this case. Finally, if $\lambda \bmod \mathfrak{q}$ is algebraic over F, then we can find a p-power q such that α^q is separable algebraic over F, see Fields, Section 037H. Note that D and $\bar{\Lambda}$ are henselian local rings, see Algebra, Lemma 06RS. Let $D \to D'$ be a finite étale extension whose residue field extension is $F(\alpha^q)/F$, see Algebra, Lemma 04GK. Since $\bar{\Lambda}$ is henselian and $F(\alpha^q)$ is contained in its residue field we can find a factorization $D' \to \bar{\Lambda}$. By the first part of the argument we see that $\bar{\lambda}^{qq'} \in D'$ for some $q' > 0$. □